

CircuitMod: Modifying The Approach To Circuit Simulation

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A dark, black screen. Symbols of a resistor, a capacitor and an inductor. A switch. Flashes of red, green and grey, and moving yellow dots. Graphs at the bottom of the screen, lines criss-crossing to create beautiful orderly curves on the screen. Click on the switch and the patterns change. The red disappears, the yellow moves differently and the grey is hardly there. Magically, a transition seems to have happened on the screen!

It does not take too much to figure out what is happening here. It certainly sounds like a circuit is kicking into action. An LCR circuit in fact. But is it possible to actually see what you can only begin to visualise? Certainly seems so! The above simply describes the window that greets you on venturing into the world of CircuitMod!

Does the scenario not sound exciting? Questions tumble into the mind one after the other. What is CircuitMod? Does it actually let you see action live? Can you make your own circuit? Does it actually mean you get to see what you have largely learnt only in theory? Let us try to seek a few answers first.

What is the big deal anyway

First things first! CircuitMod is an electronic circuit simulator. But instead of simulating a schematic in the background and giving results to you, CircuitMod shows you what is happening in the circuit. Sounds confusing? Well, let us break it down. The schematic of the circuit appears on the screen like any usual circuit diagram. But start the circuit, and a mini movie begins.

Every part of the circuit takes a

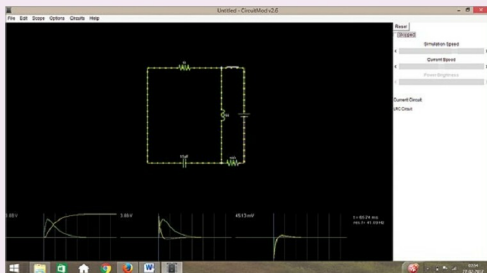


Fig. 1: A live LCR circuit with the switch closed

colour depending upon the voltage in the area. Any wire with positive voltage is seen in green, grounded regions are grey and red indicates negative voltage. Current flows as yellow dots moving on the screen.

Three graphs at the bottom of the screen act like oscilloscopes to your design, each one showing the voltage and current across a particular component. Green traces being voltage and current traces being yellow, graphs are drawn as the circuit is simulated, and you can see the variation of current and voltage as the circuit simulates over time.

Observe, change, learn

This approach seems to have a lot of takers in newbies to the field or anyone who wishes to revisit the basics of electronics. CircuitMod seems to have been constructed with this approach in mind. Go into

Quick bites

Tool: CircuitMod

Category: Electronic circuit simulation

Purpose: Enhance learning of basic electronics

Known for: Extending the capacity of the original Falstad's Java circuit simulator into CMOS chips, LED arrays, LED matrix and PIC programming

Licence: GNU general-public

Brought to you by: federoggio, munukuntla4313, fgg192

Original code by: Paul Falstad

Website: <http://circuitmod.sourceforge.net/>

the circuit menu of the tool and you have a plethora of circuits to venture into. As you surf through the circuit list, each kicks into live action, demonstrating how the circuit functions for the given set of conditions.

Now, the obvious next question is whether you only get to learn from what is already there, and the answer to that is a firm no! The pre-included circuits form the basis of most circuits an electronics beginner deals with. From then on, you are at your free will. Modify the circuit as and how you want. It could simply be changing the values of the components involved,